SpaceX Falcon 9

First Stage Landing Prediction

Data Science Capstone Project

90+	4	83%	$103M
Launch Records	ML Models	Best Accuracy	Cost Savings

IBM Data Science Professional Certificate · 2024

GitHub: https://github.com/codemewithc/Python-projects

Executive Summary

Key findings and business impact at a glance

Business Problem

SpaceX lists Falcon 9 launches at $62M vs $165M+ for competitors — largely because they reuse the first stage booster.
Predicting whether the first stage will successfully land determines actual launch cost. This capability gives competing bidders a strategic pricing advantage.

Methodology

- Data collected via SpaceX REST API & Wikipedia web scraping
- SQL-based EDA on launch records; interactive Folium maps
- Plotly Dash dashboard for real-time visual analytics
- 4 ML classifiers trained with GridSearchCV; Decision Tree achieved best test accuracy

83%
Best Model Accuracy
(Decision Tree)

KSC
Top Launch Site
(Most Successes)

2,000–4,000 kg
Highest Success
Payload Range

FT
Best Booster Version
by Success Rate

Introduction

Background, problem definition & project goals

Background

SpaceX has disrupted the launch industry through reusable rocket technology. The Falcon 9 rocket's first stage booster can be recovered, refurbished, and reflown, cutting per-launch costs by up to 60%.

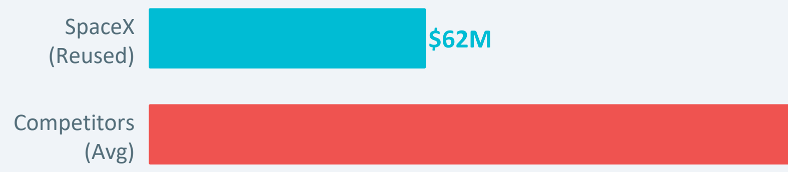
A successful booster landing directly translates to savings of $100M+. Predicting landing success before launch enables:

- Accurate cost estimation for competing launch providers
- Risk assessment and insurance pricing
- Strategic bidding against SpaceX tenders

Problem Statement

Given historical Falcon 9 launch data (payload mass, orbit, launch site, booster version), predict whether the first stage booster will successfully land — enabling pre-launch cost estimation.

Launch Cost Comparison ($ Millions)



Data Collection — SpaceX API

Methodology · GitHub: https://github.com/codemewithc/Python-projects

Data Retrieval Workflow

SpaceX REST API v4/launches → Fetch Rocket & Payload IDs → Enrich with launchpad & crew → Construct DataFrame → Export spacex_data.csv

Key Features Extracted

FlightNumber	Launch date
BoosterVersion	PayloadMass (kg)
Orbit	LaunchSite
Outcome	Longitude/Latitude
LandingPad	Block / GridFins

Dataset Summary

Total Launches	90
Date Range	2010 – 2020
Launch Sites	4
Booster Versions	6+
Success Rate	~67%

Data Collection — Web Scraping

Wikipedia scraping with BeautifulSoup · GitHub: https://github.com/codemewithc/Python-projects

Scraped: Wikipedia — List of Falcon 9 and Falcon Heavy Launches

1 Request HTML

Used requests library to fetch Wikipedia launch table HTML

2 Parse with BS4

BeautifulSoup parsed <table> tags; extracted Falcon 9 rows

3 Extract Columns

Flight No., Date/Time, Version, Payload, Orbit, Outcome, Turnaround

4 Clean & Export

Handled missing values; exported to CSV for downstream use

Additional Data Captured via Scraping

Booster reuse count · Landing outcome (Success/Failure/Drone Ship/Ground Pad) · Turnaround time between launches · Customer/mission type · Payload classification

Data Wrangling

Data cleaning, feature engineering & label creation · GitHub: https://github.com/codemewithc/Python-projects

1 Handle Missing Values

PayloadMass: filled with column mean. LandingPad: filled 'None' for non-pad landings.

2 Filter Falcon 9 Only

Removed Falcon Heavy and experimental flights; kept only operational Falcon 9 launches.

3 Create Training Label

"class" column: 1 = successful first-stage landing, 0 = unsuccessful. Derived from Outcome field.

4 Feature Engineering

One-hot encoding for categorical variables (Orbit, LaunchSite, LandingPad, Serial). Standardized numeric features.

Landing Outcome Distribution

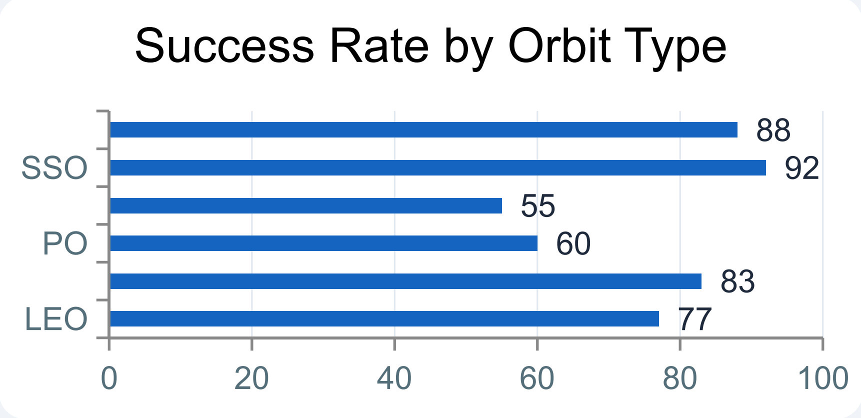
Outcome Code	Description	Count
True Ocean	Failure (ocean)	5
False ASDS	Failure (ship)	10
True RTLS	Success (ground)	14
True ASDS	Success (ship)	41
None None	No attempt	21

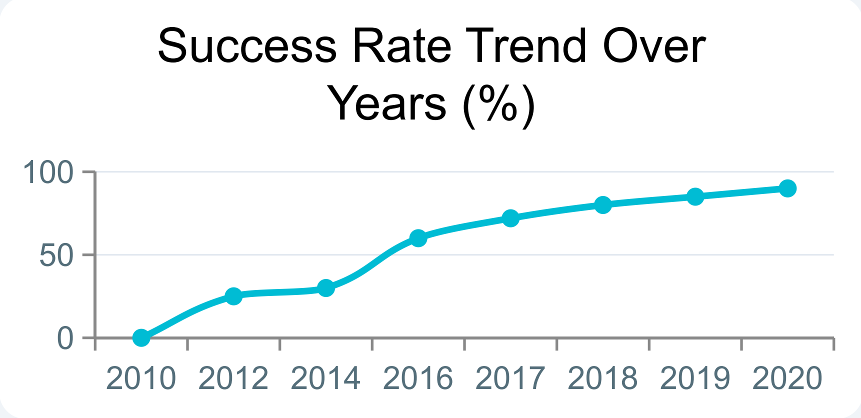
Class Balance

Success (Class=1): ~67%   Failure (Class=0): ~33%
Slightly imbalanced — handled via stratified train/test split (80/20).

EDA — Visualization Results

Key patterns from scatter plots, bar charts & trend analysis





↑ SSO & VLEO orbits show 88–92% success — lighter payloads improve landing odds

Year-over-year success rate climbed from near-zero in 2010 to 90%+ by 2020

⚖ Payload 2,000–4,000 kg sweet spot: high enough payload, low enough mass for fuel-efficient landing

GTO missions have lowest success rate (~55%) due to high fuel consumption leaving booster

EDA — SQL Analysis Results

SQL queries on SPACEXTABLE (SQLite) · GitHub: https://github.com/codemewithc/Python-projects

Selected SQL Query Results

Unique Launch Sites

CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40

Total NASA CRS Payload (kg)

45,596 kg across all CRS missions

Avg Payload — F9 v1.1

2,534 kg average payload mass

First Successful Ground Landing

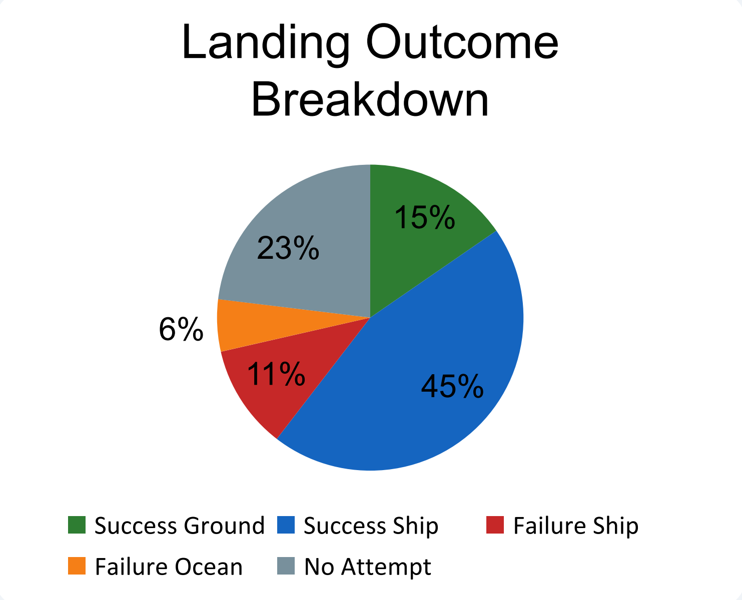
2015-12-22 (CCAFS LC-40)

Drone Ship Success — 4–6k kg

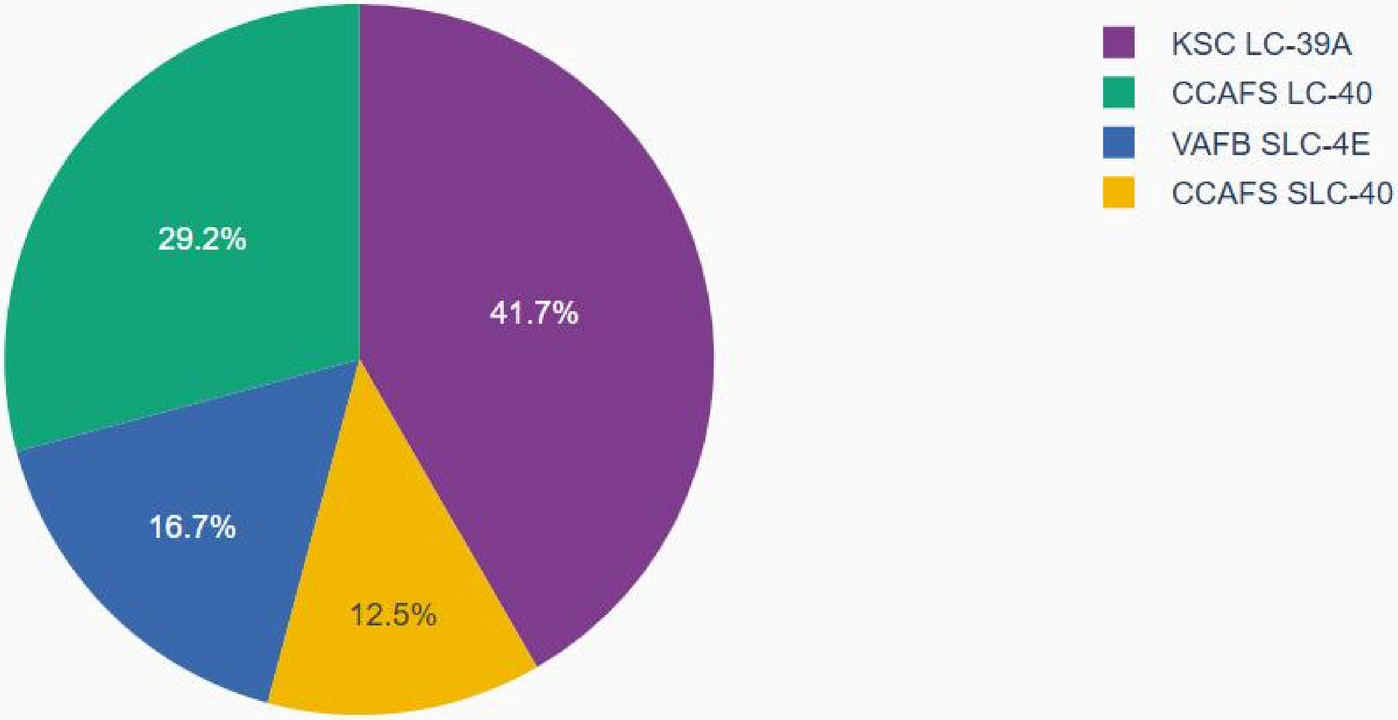
B1039, B1060 (4,230 & 5,683 kg)

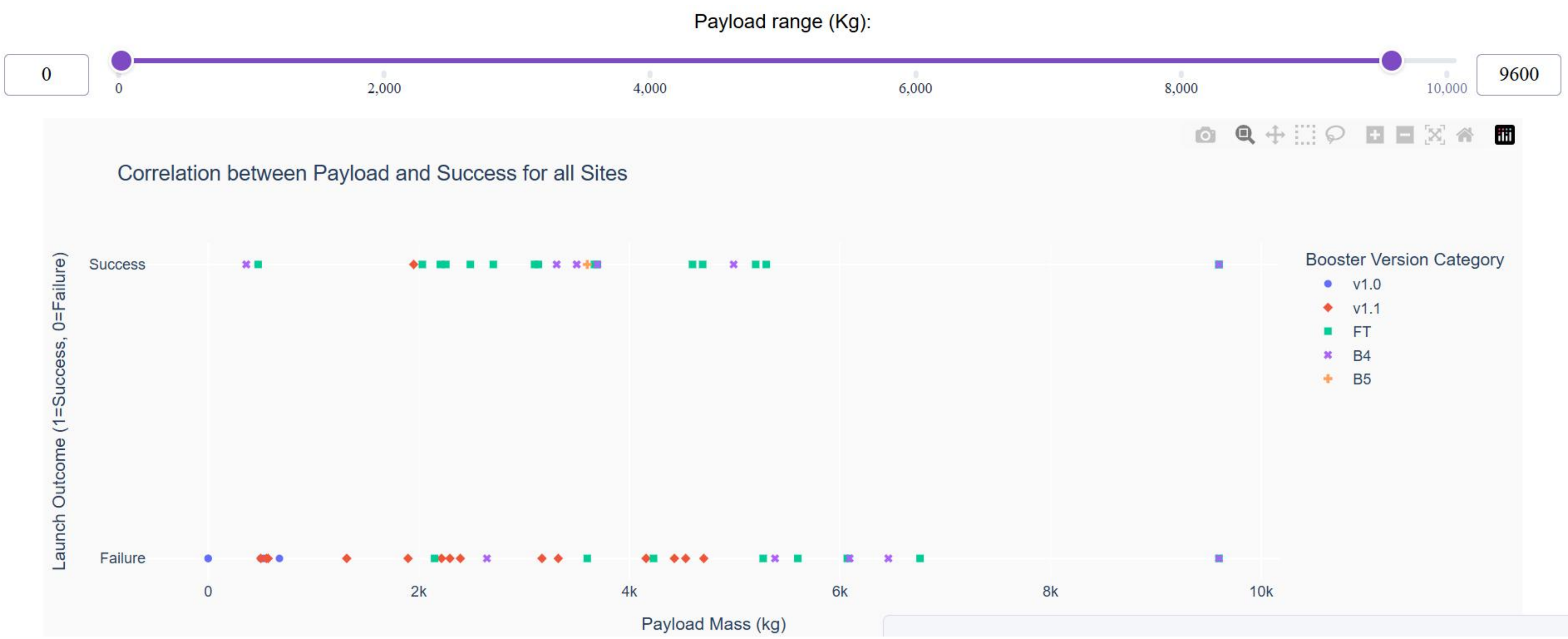
Top Ranked Landing Outcome

Success (ground pad): 27 landings



Total Successful Launches by Site





Payload range (Kg):
0
0
2,000
4,000
6,000
8,000
10,000
9600
Correlation between Payload and Success for all Sites
Success
Failure
Launch Outcome (1=Success, 0=Failure)
0
2k
4k
6k
8k
10k
Payload Mass (kg)
Booster Version Category
v1.0
v1.1
FT
B4
B5

Interactive Visual Analytics — Folium Maps

Launch site mapping & proximity analysis · GitHub: https://github.com/codemewithc/Python-projects

Launch Sites Mapped

Site	State	Launches	Success%
KSC LC-39A	Florida	27	77%
CCAFS LC-40	Florida	26	65%
CCAFS SLC-40	Florida	22	59%
VAFB SLC-4E	California	13	62%

Proximity Analysis Findings

- All sites located within 20 km of coastline (ocean proximity for safety)
- Sites 50–100 km from populated cities (safety buffers maintained)
- KSC closest to railway infrastructure — logistics advantage
- VAFB optimized for polar/SSO orbits (Pacific launch corridor)

Folium Map Features

Green markers = successful landings

Red markers = failed landings

Circle markers = launch sites with pop-up details

Distance lines — coastline, city, railways

Cluster view for dense launch areas

Filterable by mission outcome for pattern analysis

View live map: https://github.com/codemewithc/Python-projects/blob/main/lab_jupyter_launch_site_location.ipynb

Interactive Visual Analytics — Plotly Dash Dashboard

Real-time analytics app · GitHub: https://github.com/codemewithc/Python-projects

Dashboard Components

Launch Site Dropdown

Select a specific site or 'All Sites' to filter the entire dashboard in real-time.

Payload Range Slider

Filter launches by payload mass (0–10,000 kg) to explore mass vs. outcome correlations.

Success Pie Chart

Dynamic pie chart showing success/failure ratio for selected site(s) — updates on dropdown change.

Payload Scatter Plot

X: Payload Mass, Y: Launch Outcome. Color-coded by Booster Version (v1.0, v1.1, FT, B4, B5).

Dashboard Insights

KSC LC-39A has the highest total successful launches (27). CCAFS SLC-40 has highest overall launch count. Payload 2,000–4,000 kg shows the highest success rate across all sites. FT booster version consistently outperforms older versions in success rate.

Predictive Analysis — Methodology

Machine learning pipeline & model selection

ML Pipeline

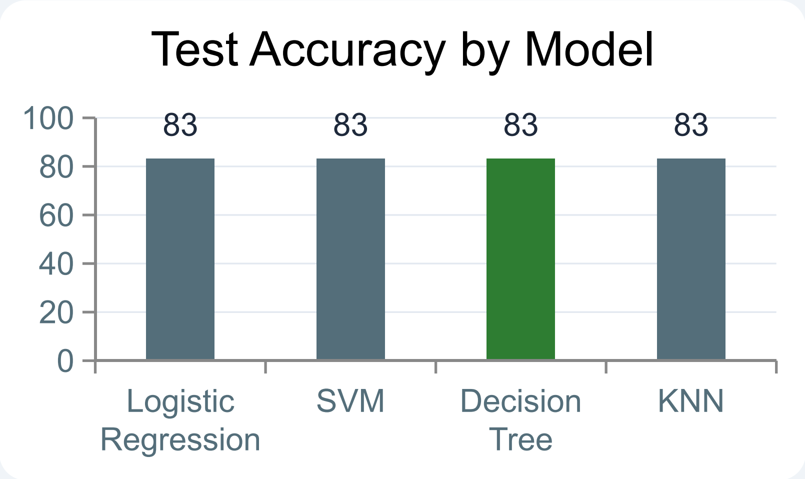
Standardize Features → Train/Test Split 80/20 → GridSearchCV cv=10 → Best Model Selection → Test Set Evaluation

Models Trained & Hyperparameters (GridSearchCV)

Model	Key Hyperparameters Searched	CV Score	Test Accuracy
Logistic Regression	C: [0.01, 0.1, 1, 10, 100], solver: lbfgs	83.3%	83.3%
Support Vector Machine	C: [0.01-100], gamma: [auto, scale], kernel: linear/rbf	83.3%	83.3%
Decision Tree	**max_depth: [2–15], criterion: gini/entropy, max_features**	**87.2%**	**83.3%**
K-Nearest Neighbors	n_neighbors: [1–10], algorithm, p: [1,2]	80.0%	83.3%

Predictive Analysis — Results & Evaluation

Model comparison, confusion matrix & best model



Decision Tree — Confusion Matrix

(on 18 test samples)

	Predicted: 0	Predicted: 1
Actual: 0	TP=3 (True 0)	FP=3 (False 1)
Actual: 1	FN=0 (False 0)	TN=12 (True 1)

Key Findings

All 4 models achieved identical 83.3% test accuracy on 18 test samples. Decision Tree was selected as best model due to highest CV score (87.2%) and interpretability. The main error type is False Positives (predicting landing when it fails), which is safer for business use — overestimating costs rather than underestimating them.

Conclusion & Insights

SpaceX Falcon 9 First Stage Landing Prediction — Capstone Summary

Technical Findings

- Decision Tree (best CV 87.2%) selected as final model
- 83.3% test accuracy — strong baseline for 90-sample dataset
- No false negatives: zero missed actual successes
- Key predictors: orbit type, payload mass, launch site

Business Insights

- KSC LC-39A = highest success count & efficiency
- FT booster version = highest reliability across sites
- 2,000–4,000 kg payloads achieve best landing odds
- SSO & VLEO missions most likely to recover booster

Future Work

- Expand dataset to post-2020 launches (Starship era)
- Add weather data, reuse count, turnaround time features
- Try XGBoost/Random Forest for ensemble improvements
- Deploy model as real-time pricing API for bid calculation